

Learner Segmentation API

API Reference

Base URL

`https://bachhaovan1906-it5135e.hf.space`

All responses are JSON. All requests and responses use UTF-8.

Method	Endpoint	Purpose
GET	/	Health check
GET	/clusters	List all clusters and their profiles
POST	/predict	Assign one learner to a segment

1. Health Check

GET /

Confirms the API is running and the model is loaded. Takes no input.

Example response

```
{
  "status": "ok",
  "n_clusters": 6,
  "n_features": 39
}
```

2. List Clusters

GET /clusters

Returns an array with one object per cluster. Takes no input. Use this to display the behavioral profiles of all learner segments.

Response fields (per cluster)

Field	Type	Valid range / values	Description
cluster_id	integer	0 to n_clusters-1	Numeric ID of the cluster
persona	string	-	Human-readable persona name
risk_level	string	Low / Medium / High	Risk category for the persona
description	string	-	What characterizes this segment
recommendation	string	-	Suggested action for this segment
feature_means	object	-	Average value of each of the 39 features in this cluster

Example response

```
[
  {
    "cluster_id": 0,
```

```
"persona": "Consistent Achiever",
"risk_level": "Low",
"description": "High engagement and strong performance.",
"recommendation": "Offer enrichment and advanced tasks.",
"feature_means": {
  "login_weekly": 7.21,
  "video_completion_rate": 0.88,
  "forum_posts_count": 4.6,
  "...": "... all 39 features"
}
},
{ "cluster_id": 1, "persona": "At-Risk Disengaged", "...": "..." }
]
```

3. Predict Learner Segment

POST /predict

Send one learner's raw activity data as a JSON body. The API computes the model features internally and returns the assigned segment. Send the 6 required fields at minimum; include optional fields for a more accurate result (omitted optional fields fall back to training-data defaults).

Request body - required fields

Field	Type	Valid range / values	Description
login_weekly	number	≥ 0	Average logins per week
dominant_login_time	string	Morning / Afternoon / Evening / Night	Usual time of day the learner studies
video_completion_rate	number	0.0 to 1.0	Fraction of course videos completed
days_to_first_activity	number or null	≥ 0 , or null	Days from enrollment to first activity; null = never started
submission_timeliness_days	number	any (+/-)	Avg submission timing; negative = early, positive = late
forum_posts_count	integer	≥ 0	Total forum posts

Request body - optional fields

Field	Type	Valid range / values	Description
date_registration	number	any (+/-)	Days from course start (negative = before start)
code_module	string	e.g. AAA, BBB	Course module code
code_presentation	string	e.g. 2013J	Course presentation code
vle_total_clicks	number	≥ 0	Total platform clicks
vle_active_days	number	≥ 0	Distinct days with activity
vle_distinct_sites	number	≥ 0	Distinct resources visited
vle_clicks_forumng	number	≥ 0	Clicks on the forum
vle_clicks_oucontent	number	≥ 0	Clicks on content pages
vle_clicks_quiz	number	≥ 0	Clicks on quizzes
vle_clicks_homepage	number	≥ 0	Clicks on the homepage
vle_clicks_resource	number	≥ 0	Clicks on resources
vle_clicks_subpage	number	≥ 0	Clicks on subpages
vle_clicks_url	number	≥ 0	Clicks on URLs
assess_n_submitted	number	≥ 0	Number of assessments submitted
assess_mean_score	number	0 to 100	Average assessment score
assess_n_banked	number	≥ 0	Number of banked assessments
assess_mean_days_before_deadline	number	any (+/-)	Avg days submitted before deadline
assess_n_late	number	≥ 0	Number of late submissions

Note: dominant_login_time must be exactly one of the four allowed words (capitalized). Any other value returns a 422 validation error.

Example request - full

POST /predict
Content-Type: application/json

```
{
  "login_weekly": 8,
  "dominant_login_time": "Morning",
  "video_completion_rate": 0.9,
  "days_to_first_activity": 3,
  "submission_timeliness_days": -1.5,
  "forum_posts_count": 5,
  "date_registration": -20,
  "code_module": "AAA",
  "code_presentation": "2013J",
  "vle_total_clicks": 450,
  "vle_active_days": 40,
  "vle_distinct_sites": 25,
  "vle_clicks_forumng": 60,
  "vle_clicks_oucontent": 120,
  "vle_clicks_quiz": 80,
  "vle_clicks_homepage": 90,
  "vle_clicks_resource": 50,
  "vle_clicks_subpage": 30,
  "vle_clicks_url": 20,
  "assess_n_submitted": 5,
  "assess_mean_score": 82,
  "assess_n_banked": 0,
  "assess_mean_days_before_deadline": 4,
  "assess_n_late": 0
}
```

Example request - minimal (only required fields)

POST /predict
Content-Type: application/json

```
{
  "login_weekly": 1,
  "dominant_login_time": "Night",
  "video_completion_rate": 0.1,
  "days_to_first_activity": null,
  "submission_timeliness_days": 9,
  "forum_posts_count": 0
}
```

Example response

Field	Type	Valid range / values	Description
cluster_id	integer	0 to n_clusters-1	Assigned cluster ID
persona	string	-	Assigned persona name
risk_level	string	Low / Medium / High	Risk category
description	string	-	What characterizes this segment
recommendation	string	-	Suggested action
engineered_features	object	-	The 39 computed features (for reference; can be ignored)

```
{
  "cluster_id": 0,
  "persona": "Consistent Achiever",
  "risk_level": "Low",
  "description": "High engagement and strong performance.",
  "recommendation": "Offer enrichment and advanced tasks.",
  "engineered_features": {}
}
```

```
"engineered_features": {  
  "login_weekly": 8.0,  
  "frequency_score": 1.83,  
  "...": "... all 39 features"  
}  
}
```

4. Error Responses

Status	Meaning	Common cause
200	Success	Request processed normally
422	Validation error	Missing required field, wrong type, or invalid dominant_login_time value. Response body lists the
500	Server error	Prediction failed internally. Contact the API owner.

Example 422 response

```
{  
  "detail": [  
    {  
      "loc": ["body", "video_completion_rate"],  
      "msg": "Input should be less than or equal to 1",  
      "type": "less_than_equal"  
    }  
  ]  
}
```

Interactive documentation: append /docs to the base URL in a browser to explore and test every endpoint live.